

top 10 genus

Top 10 genus

library prep

install

activate

```
library(readxl)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(ggplot2)
library(tibble)
library(vegan)
```

Loading required package: permute

```
library(knitr)
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
library(stringr)
library(forcats)
library(purrr)
library(patchwork)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v readr 2.1.5
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag() masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(broom)
library(ggvenn)
library(grid)
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:readr':

col_factor

The following object is masked from 'package:purrr':

discard

Retrieve data

Home PC

```
# microscopy
```

```
R_5_abundance_2023 <- read_excel(  
  "C:/Program Files (x86)/Results/Planteplanktontellinger/R-5 2023/R-5_Ringdalsfjorden_kar  
  na = "."  
)
```

New names:

```
* `` -> `...1`
```

```
R_5_abundance_2024 <- read_excel(  
  "C:/Program Files (x86)/Results/Planteplanktontellinger/R-5 2024/R-5_Ringdalsfjorden_kar  
  na = "."  
)
```

New names:

```
* `` -> `...1`
```

```
S_9_abundance_2023 <- read_excel(  
  "C:/Program Files (x86)/Results/Planteplanktontellinger/S-9 2023/S-9_Haslau_karbon_2023.  
  na = "."  
)
```

New names:

```
* `` -> `...1`
```

```
S_9_abundance_2024 <- read_excel(  
  "C:/Program Files (x86)/Results/Planteplanktontellinger/S-9 2024/S-9_Haslau_karbon_2024.  
  na = "."  
)
```

New names:

```
* `` -> `...1`
```

```
#metabarcoding
setwd("C:/Results")

metadata <- read_excel("samples Oslo Fjord.xlsx")

Metabarcoding_results <- read_excel("Oslofjord_illumina_dada2.xlsx")
```

School PC

Data cleaning

Metabarcoding

```
# Keep only rows with the station IDs i need and all negatives
metadata <- subset(metadata, station_id %in% c("R-5", "S-9", "Negative"))

# remove the negatives i dont need
metadata <- subset(metadata, !(file_name_R1 %in% c("14-OS24_S14_L001_R1_001.fastq.gz", "21-OS24_S14_L001_R1_001.fastq.gz")))

cols_to_remove <- c(
  "1-OS51", "2-OS52", "3-OS57", "4-OS23", "5-OS46", "6-OS30", "7-OS50",
  "8-OS17", "9-OS18", "10-OS19", "11-OS20", "12-OS21", "13-OS22", "14-OS24",
  "15-OS25", "16-OS26", "17-OS27", "18-OS28", "19-OS29", "20-OS31", "21-OS32",
  "22-OS41", "23-OS43", "24-OS44", "25-OS45", "26-OS47", "27-OS48", "28-OS49",
  "29-OS53", "30-OS54", "31-OS55", "32-OS56", "33-OS42", "34-OS44_2",
  "35-OS45_2", "36-OS47_2", "140-S28", "141-S29", "120-S8", "128-S16", "136-S24", "142-S30"
)

Metabarcoding_results <- Metabarcoding_results %>%
  select(-any_of(cols_to_remove))

library(dplyr)

metabarcoding_filtered <- Metabarcoding_results %>%
  mutate(class = recode(as.character(class),
    "Bacillariophyceae" = "Diatoms",
```

```

      "Mediophyceae"          = "Diatoms",
      "Coscinodiscophyceae"   = "Diatoms",
      .default = as.character(class)))

# removing everything but diatoms and dinoflagellata
metabarcoding_filtered <- metabarcoding_filtered %>%
  mutate(
    class_clean = ifelse(is.na(class), NA_character_, str_trim(as.character(class))),
    subdivision_clean = ifelse(is.na(subdivision), NA_character_, str_trim(as.character(subdivision)))
  ) %>%
  filter(
    # keep rows where class == "Diatoms" (case-insensitive)
    (!is.na(class_clean) & tolower(class_clean) == "diatoms") |
    # or where subdivision == "Dinoflagellata" (case-insensitive)
    (!is.na(subdivision_clean) & tolower(subdivision_clean) == "dinoflagellata")
  ) %>%
  # drop the helper clean columns (optional)
  select(-class_clean, -subdivision_clean)

# remove samples with a bootstrap value below 80 at subdivision
metabarcoding_filtered <- metabarcoding_filtered[metabarcoding_filtered$subdivision_boot >= 80, ]

metabarcoding_filtered <- metabarcoding_filtered %>%
  filter(class != "Syndiniales" | is.na(class))

cols_to_remove <- c(
  "asv_code", "domain", "supergroup", "division", "order", "family", "species",
  "domain_boot", "supergroup_boot", "division_boot", "order_boot", "family_boot",
  "genus_boot", "subdivision_boot", "class_boot", "species_boot", "asv_number", "sequence_id"
)

metabarcoding_filtered <- metabarcoding_filtered %>%
  select(-any_of(cols_to_remove))
library(dplyr)

metabarcoding_filtered <- metabarcoding_filtered %>%
  mutate(subdivision = if_else(subdivision == "Gyrista", "Diatoms", subdivision))

```

```

metabarcoding_filtered <- metabarcoding_filtered %>%
  mutate(
    genus = case_when(
      genus %in% c("Gyrodinium", "Gymnodinium") ~ "Gymnodiniales",
      TRUE ~ genus
    )
  )

# Define R-5 columns:
r5_cols <- c("113-S1", "114-S2", "115-S3", "116-S4",
            "117-S5", "118-S6", "119-S7", "121-S9",
            "122-S10", "123-S11")

# splitting to R-5 and S-9

metabarcoding_R5 <- metabarcoding_filtered %>%
  select(subdivision, genus, all_of(r5_cols))

metabarcoding_S9 <- metabarcoding_filtered %>%
  select(-all_of(r5_cols))

## R-5: split into Diatoms and Dinoflagellata
metabarcoding_R5_diatoms <- metabarcoding_R5 %>%
  filter(subdivision == "Diatoms")

metabarcoding_R5_dino <- metabarcoding_R5 %>%
  filter(subdivision == "Dinoflagellata")

## S-9: split into Diatoms and Dinoflagellata
metabarcoding_S9_diatoms <- metabarcoding_S9 %>%
  filter(subdivision == "Diatoms")

metabarcoding_S9_dino <- metabarcoding_S9 %>%
  filter(subdivision == "Dinoflagellata")

#combines the genus

combine_by_genus <- function(df) {
  df %>%
    group_by(genus) %>%
    summarise(

```

```

    across(where(is.numeric), ~ sum(.x, na.rm = TRUE)),
    .groups = "drop"
  )
}
metabarcoding_R5_diatoms <- combine_by_genus(metabarcoding_R5_diatoms)
metabarcoding_R5_dino <- combine_by_genus(metabarcoding_R5_dino)
metabarcoding_S9_diatoms <- combine_by_genus(metabarcoding_S9_diatoms)
metabarcoding_S9_dino <- combine_by_genus(metabarcoding_S9_dino)

#change to %
to_column_percent <- function(df) {
  df %>%
    mutate(
      across(
        .cols = where(is.numeric), # all numeric columns
        .fns = ~ .x / sum(.x, na.rm = TRUE) * 100
      )
    )
}

metabarcoding_R5_diatoms <- to_column_percent(metabarcoding_R5_diatoms)
metabarcoding_R5_dino <- to_column_percent(metabarcoding_R5_dino)
metabarcoding_S9_diatoms <- to_column_percent(metabarcoding_S9_diatoms)
metabarcoding_S9_dino <- to_column_percent(metabarcoding_S9_dino)

#make top 10 genus lists

keep_top10_and_other <- function(df) {
  # 1. Add total abundance per genus
  df_with_total <- df %>%
    mutate(
      total_abundance = rowSums(across(where(is.numeric)))
    )

  # 2. Top 10 genera
  top10 <- df_with_total %>%
    arrange(desc(total_abundance)) %>%
    slice_head(n = 10)
}

```

```

top10_genera <- top10$genus

# 3. "Other" = sum of all remaining genera
other_row <- df_with_total %>%
  filter(!genus %in% top10_genera) %>%
  summarise(
    genus      = "Other",
    across(where(is.numeric), ~ sum(.x, na.rm = TRUE))
  )

# 4. Combine and drop helper column
result <- top10 %>%
  bind_rows(other_row) %>%
  select(-total_abundance)

  result
}

R5_diatoms_top10 <- keep_top10_and_other(metabarcoding_R5_diatoms)
R5_dino_top10    <- keep_top10_and_other(metabarcoding_R5_dino)
S9_diatoms_top10 <- keep_top10_and_other(metabarcoding_S9_diatoms)
S9_dino_top10    <- keep_top10_and_other(metabarcoding_S9_dino)

#pivot to long format

to_long <- function(df) {
  df %>%
    pivot_longer(
      cols      = -genus,          # all columns except genus
      names_to  = "Sample",        # new column with old column names
      values_to = "relative_abundance" # new column with the values
    )
}

R5_diatoms_long <- to_long(R5_diatoms_top10)
R5_dino_long    <- to_long(R5_dino_top10)
S9_diatoms_long <- to_long(S9_diatoms_top10)
S9_dino_long    <- to_long(S9_dino_top10)

#clean metadata and insert dates to the long format datasets

```



```

metadata_clean <- metadata %>%
  # rename file_name_R1 (case-insensitive) to Sample if present
  rename_with(.fn = ~ "Sample", .cols = matches("(?i)^file_name[_-]?r1$")) %>%
  # ensure Sample exists
  { if (!"Sample" %in% names(.)) stop("Could not find a column matching 'file_name_R1' in
  # trim whitespace and coerce to character
  mutate(Sample = str_trim(as.character(Sample))) %>%
  # keep only the part before the first underscore (e.g., "113-S1_..." -> "113-S1")
  mutate(Sample = ifelse(str_detect(Sample, "_"),
    str_replace(Sample, "^(.*)_.*$", "\\1"),
    Sample)) %>%
  # decide which columns to keep: Sample, station_id (if present), and first date-like col
  {
    df <- .
    date_cols <- names(df)[str_detect(names(df), regex("date", ignore_case = TRUE))]
    has_station <- "station_id" %in% names(df)

    if (length(date_cols) == 0) {
      # no date column found
      if (has_station) {
        df2 <- df %>% select(Sample, station_id)
      } else {
        df2 <- df %>% select(Sample)
      }
    } else {
      # keep first date-like column and station_id if present
      date_col <- date_cols[1]
      if (has_station) {
        df2 <- df %>% select(Sample, all_of(date_col), station_id) %>%
          rename(Date = all_of(date_col))
      } else {
        df2 <- df %>% select(Sample, all_of(date_col)) %>%
          rename(Date = all_of(date_col))
      }
    }
    df2
  } %>%
  # final trim and remove empty Sample rows
  mutate(Sample = str_trim(as.character(Sample))) %>%
  filter(!(is.na(Sample) | Sample == ""))

```

```

# Ensure Sample column exists in metadata_clean and is character/trimmed
metadata_clean <- metadata_clean %>%
  mutate(Sample = as.character(Sample), Sample = str_trim(Sample))

# Helper: safe join a metabarcoding long table to metadata
join_with_meta <- function(mb_long, meta = metadata_clean, join_type = "left",
                           sample_col = "Sample") {
  # normalize sample column on metabarcoding table
  mb <- mb_long %>%
    mutate(!sample_col := as.character(.data[[sample_col]]),
           !sample_col := str_trim(.data[[sample_col]]))

  # choose join function
  joined <- switch(join_type,
                   left = left_join(mb, meta, by = sample_col),
                   inner = inner_join(mb, meta, by = sample_col),
                   stop("join_type must be 'left' or 'inner'"))

  # return joined table
  joined
}

# Join each metabarcoding table
S9_diatoms_long <- join_with_meta(S9_diatoms_long, metadata_clean, join_type = "left")

R5_diatoms_long <- join_with_meta(R5_diatoms_long, metadata_clean, join_type = "left")

S9_dino_long <- join_with_meta(S9_dino_long, metadata_clean, join_type = "left")

R5_dino_long <- join_with_meta(R5_dino_long, metadata_clean, join_type = "left")

```

microscopy

```

clean_abundance_data <- function(df) {
  df %>%
    # rename first two columns
    rename(higher_taxa = 1, Species = 2) %>%

    # remove rows where both columns are empty

```

```

filter(!(is.na(higher_taxa) & is.na(Species))) %>%

# fill down higher_taxa names
fill(higher_taxa, .direction = "down") %>%

# remove rows without species
filter(!is.na(Species) & Species != "") %>%

# remove summary rows
filter(!(Species %in% c("Sum:", "Sum total:"))) %>%

# convert "." to NA if any remain (for safety)
mutate(across(everything(), ~na_if(., "."))) %>%

# convert numeric columns to numeric if read as character
mutate(across(-c(higher_taxa, Species), as.numeric)) %>%

# convert abundance values to relative abundance (%)
mutate(across(-c(higher_taxa, Species),
               ~ (. / sum(., na.rm = TRUE)) * 100))
}

# Apply to your datasets
R_5_abundance_2023_clean <- clean_abundance_data(R_5_abundance_2023)
R_5_abundance_2024_clean <- clean_abundance_data(R_5_abundance_2024)
S_9_abundance_2023_clean <- clean_abundance_data(S_9_abundance_2023)
S_9_abundance_2024_clean <- clean_abundance_data(S_9_abundance_2024)

# removing unusable letters
# data frame names to process
df_names <- c(
  "R_5_abundance_2023_clean",
  "R_5_abundance_2024_clean",
  "S_9_abundance_2023_clean",
  "S_9_abundance_2024_clean"
)

# target keywords and their canonical replacements
replacements <- c(
  chlorophyta = "Chlorophyta",

```

```

euglenophyceae = "Euglenophyceae",
cyanobacteria  = "Cyanobacteria",
prasinophyceae = "Prasinophyceae"
)

for (nm in df_names) {
  if (!exists(nm, envir = .GlobalEnv)) {
    warning("Object '", nm, "' not found; skipping.")
    next
  }
  df <- get(nm, envir = .GlobalEnv)
  if (!"higher_taxa" %in% names(df)) {
    warning("Data frame '", nm, "' has no column 'higher_taxa'; skipping.")
    next
  }

  # ensure character and trim
  df$higher_taxa <- trimws(as.character(df$higher_taxa))

  # for each keyword, find rows containing it and set to canonical form
  for (kw in names(replacements)) {
    mask <- grepl(kw, df$higher_taxa, ignore.case = TRUE, perl = TRUE)
    if (any(mask)) df$higher_taxa[mask] <- replacements[[kw]]
  }

  # tidy whitespace
  df$higher_taxa <- gsub("\\s+", " ", df$higher_taxa)
  df$higher_taxa <- trimws(df$higher_taxa)

  assign(nm, df, envir = .GlobalEnv)
  message(sprintf("Processed %s: replaced %d rows matching targets.",
                  nm, sum(grepl(paste(names(replacements), collapse="|"), df$higher_taxa,

```

Processed R_5_abundance_2023_clean: replaced 21 rows matching targets.

Processed R_5_abundance_2024_clean: replaced 14 rows matching targets.

Processed S_9_abundance_2023_clean: replaced 11 rows matching targets.

Processed S_9_abundance_2024_clean: replaced 13 rows matching targets.

```

# named mapping: names = old values, values = new values
mapping <- c(
  "Bacillariophyceae (kiselalger)" = "Diatoms",
  "Chlorophyta" = "Chlorophyta",
  "Choanoflagellata (krageflagellater)" = "Other Protists",
  "Chrysophyceae (gullalger)" = "Other Ochrophyta",
  "Ciliophora (ciliater)" = "Ciliophora",
  "Coccolithophyceae (kalk- og svepeflagellater)" = "Haptophyta",
  "Conjugatophyceae" = "Chlorophyta",
  "Cryptophyceae (svelgflagellater)" = "Cryptophyta",
  "Dictyochophyceae (kiselflagellater og pedineller)" = "Other Ochrophyta",
  "Dinophyceae (fureflagellater)" = "Dinophyta",
  "Ebriophyceae (skjelettflagellater)" = "Other Protists",
  "Euglenophyceae" = "Euglenophyta",
  "Imbricatea" = "Other Protists",
  "Prasinophyceae" = "Chlorophyta",
  "Pyramimonadophyceae" = "Chlorophyta",
  "Synurophyceae" = "Other Ochrophyta",
  "Telonemea" = "Other Protists"
)
mapping2 <- c(
  "Scrippsiella-gruppen" = "Scrippsiella"
)
# rows to drop
drop_these <- c(
  "Cyanobacteria",
  "Classes incertae sedis (ubestemte klasser)"
)

# put your datasets in a named list
dfs <- list(
  R_5_abundance_2023_clean = R_5_abundance_2023_clean,
  R_5_abundance_2024_clean = R_5_abundance_2024_clean,
  S_9_abundance_2023_clean = S_9_abundance_2023_clean,
  S_9_abundance_2024_clean = S_9_abundance_2024_clean
)

# function to remap and filter one dataframe
fix_taxa <- function(df) {
  df %>%
    # make sure higher_taxa is character to map reliably

```

```

mutate(higher_taxa = as.character(higher_taxa)) %>%
# replace only exact matches listed in mapping
mutate(higher_taxa = dplyr::recode(higher_taxa, !!!mapping)) %>%
  mutate(Species = dplyr::recode(Species, !!!mapping2)) %>%
# remove unwanted groups
filter(!higher_taxa %in% drop_these)
}

# apply to all and reassign back to the global env (or keep as a list)
fixed_list <- lapply(dfs, fix_taxa)

# if you want to overwrite the original objects in the global environment:
list2env(fixed_list, envir = .GlobalEnv)

```

<environment: R_GlobalEnv>

```

#fix relative abundance after the removal of certain groups
# convert abundance values to relative abundance (%)
relative_abundance <- function(df) {
  df %>%
mutate(across(-c(higher_taxa, Species),
               ~ (. / sum(., na.rm = TRUE)) * 100)))

R_5_abundance_2023_clean <- relative_abundance(R_5_abundance_2023_clean)
R_5_abundance_2024_clean <- relative_abundance(R_5_abundance_2024_clean)
S_9_abundance_2023_clean <- relative_abundance(S_9_abundance_2023_clean)
S_9_abundance_2024_clean <- relative_abundance(S_9_abundance_2024_clean)

# detect key column names (case-insensitive)
detect_and_rename_keys <- function(df) {
  nm <- names(df)
  higher_idx <- grep("higher", nm, ignore.case = TRUE)
  species_idx <- grep("species", nm, ignore.case = TRUE)
  if (length(higher_idx) < 1 || length(species_idx) < 1) {
    stop("Could not detect both higher_taxa and Species columns in the dataframe. Found: "
         paste(nm, collapse = ", "))
  }
  df %>%
    rename(
      higher_taxa = !!sym(nm[higher_idx[1]]),

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    Species = !!sym(nm[species_idx[1]])
  )
}

prepare_df_explicit <- function(df) {
  df2 <- detect_and_rename_keys(df)
  # explicitly list sample columns (everything except the two keys)
  sample_cols <- setdiff(names(df2), c("higher_taxa", "Species"))
  df2 %>%
    # make keys character
    mutate(across(all_of(c("higher_taxa", "Species")), as.character)) %>%
    # coerce sample cols to numeric
    mutate(across(all_of(sample_cols), ~ as.numeric(.))) %>%
    # group and sum duplicates within-year if present
    group_by(higher_taxa, Species) %>%
    summarise(across(all_of(sample_cols), ~ sum(.x, na.rm = TRUE)), .groups = "drop")
}

# S-9 2024
S_9_abundance_2024_clean$Species <- as.character(S_9_abundance_2024_clean$Species)
S_9_abundance_2024_clean$Species <- gsub("^cf\\.\\.\\s*", "", S_9_abundance_2024_clean$Species)

# S-9 2023
S_9_abundance_2023_clean$Species <- as.character(S_9_abundance_2023_clean$Species)
S_9_abundance_2023_clean$Species <- gsub("^cf\\.\\.\\s*", "", S_9_abundance_2023_clean$Species)

# R-5 2024
R_5_abundance_2024_clean$Species <- as.character(R_5_abundance_2024_clean$Species)
R_5_abundance_2024_clean$Species <- gsub("^cf\\.\\.\\s*", "", R_5_abundance_2024_clean$Species)

# R-5 2023
R_5_abundance_2023_clean$Species <- as.character(R_5_abundance_2023_clean$Species)
R_5_abundance_2023_clean$Species <- gsub("^cf\\.\\.\\s*", "", R_5_abundance_2023_clean$Species)

R_5_abundance_2023_clean$Species <- sub(
  "Sentriske", "Centric", R_5_abundance_2023_clean$Species
)

```

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R_5_abundance_2024_clean$Species <- sub(
  "Sentriske", "Centric", R_5_abundance_2024_clean$Species
)

S_9_abundance_2023_clean$Species <- sub(
  "Sentriske", "Centric", S_9_abundance_2023_clean$Species
)

S_9_abundance_2024_clean$Species <- sub(
  "Sentriske", "Centric", S_9_abundance_2024_clean$Species
)

combine_wide_explicit <- function(df1, df2) {
  df1p <- prepare_df_explicit(df1)
  df2p <- prepare_df_explicit(df2)
  full_join(df1p, df2p, by = c("higher_taxa", "Species"))
}

# run the combines
R_5_combined_wide <- combine_wide_explicit(R_5_abundance_2023_clean, R_5_abundance_2024_clean)
S_9_combined_wide <- combine_wide_explicit(S_9_abundance_2023_clean, S_9_abundance_2024_clean)

# optional: replace NA with 0
R_5_combined_wide[is.na(R_5_combined_wide)] <- 0
S_9_combined_wide[is.na(S_9_combined_wide)] <- 0

# example: R_5_combined_wide (assumes you already created this)
sample_cols_wide <- setdiff(names(R_5_combined_wide), c("higher_taxa", "Species"))

R_5_combined <- R_5_combined_wide %>%
  # ensure numeric for sample cols
  mutate(across(all_of(sample_cols_wide), ~ as.numeric(.))) %>%
  # convert to relative % per column (sample)
  mutate(across(all_of(sample_cols_wide),
    ~ if (sum(., na.rm = TRUE) == 0) {
      # if column sums to zero, leave as 0 (or NA if you prefer)
      rep(0, length(.))
    } else {
      . / sum(., na.rm = TRUE) * 100
    }
  ))

```



```

    ))) %>%
# optional: keep NA where original was NA instead of replacing with 0
# If you prefer NAs to remain, remove the explicit zero replacement above
ungroup()

sample_cols_s9 <- setdiff(names(S_9_combined_wide), c("higher_taxa", "Species"))
S_9_combined <- S_9_combined_wide %>%
  mutate(across(all_of(sample_cols_s9), ~ as.numeric(.))) %>%
  mutate(across(all_of(sample_cols_s9),
    ~ if (sum(., na.rm = TRUE) == 0) rep(0, length(.)) else . / sum(., na.rm =

# removing everything except diatoms and dinophyta

keep_groups <- c("diatoms", "dinophyta")

filter_higher_wide <- function(df) {
  sample_cols <- setdiff(names(df), c("higher_taxa", "Species"))
  df %>%
    mutate(higher_taxa = trimws(as.character(higher_taxa))) %>%
    filter(tolower(higher_taxa) %in% keep_groups) %>%
    # ensure sample columns numeric
    mutate(across(all_of(sample_cols), ~ as.numeric(.)))
}

R_5_diatom_dino_wide <- filter_higher_wide(R_5_combined_wide)
S_9_diatom_dino_wide <- filter_higher_wide(S_9_combined_wide)

# helper to filter by higher_taxa (robust to case/whitespace)
filter_higher_wide <- function(df, keep_group) {
  sample_cols <- setdiff(names(df), c("higher_taxa", "Species"))
  df %>%
    mutate(higher_taxa = trimws(as.character(higher_taxa))) %>%
    filter(tolower(higher_taxa) == tolower(keep_group)) %>%
    # ensure the sample columns are numeric
    mutate(across(all_of(sample_cols), ~ as.numeric(.)))
}

# create wide datasets
R_5_diatoms_wide <- filter_higher_wide(R_5_combined_wide, "Diatoms")
R_5_dinophyta_wide <- filter_higher_wide(R_5_combined_wide, "Dinophyta")

```

```

S_9_diatoms_wide    <- filter_higher_wide(S_9_combined_wide, "Diatoms")
S_9_dinophyta_wide <- filter_higher_wide(S_9_combined_wide, "Dinophyta")

#recalculate relative abundance

renormalize_wide <- function(df) {
  sample_cols <- setdiff(names(df), c("higher_taxa", "Species"))
  df %>%
    mutate(across(all_of(sample_cols),
      ~ if (sum(., na.rm = TRUE) == 0) 0 else . / sum(., na.rm = TRUE) * 100))
}

R_5_diatoms_wide_rel    <- renormalize_wide(R_5_diatoms_wide)
R_5_dinophyta_wide_rel  <- renormalize_wide(R_5_dinophyta_wide)
S_9_diatoms_wide_rel    <- renormalize_wide(S_9_diatoms_wide)
S_9_dinophyta_wide_rel  <- renormalize_wide(S_9_dinophyta_wide)

#convert to genus not species

clean_genus <- function(df) {
  df %>%
    rename(genus = Species) %>%          # 1) rename column
    mutate(
      genus = as.character(genus),
      genus = sub(" .*", "", genus)      # 2) keep everything up to first space
    )
}

R_5_diatoms_wide_rel    <- clean_genus(R_5_diatoms_wide_rel)
R_5_dinophyta_wide_rel  <- clean_genus(R_5_dinophyta_wide_rel)
S_9_diatoms_wide_rel    <- clean_genus(S_9_diatoms_wide_rel)
S_9_dinophyta_wide_rel  <- clean_genus(S_9_dinophyta_wide_rel)

# rename the species column to genus
combine_by_genus <- function(df) {
  df %>%
    group_by(genus) %>%
    summarise(
      across(where(is.numeric), ~ sum(.x, na.rm = TRUE)),
      .groups = "drop"
    )
}

```

```

}

R_5_diatoms_wide_rel <- combine_by_genus(R_5_diatoms_wide_rel)
R_5_dinophyta_wide_rel <- combine_by_genus(R_5_dinophyta_wide_rel)
S_9_diatoms_wide_rel <- combine_by_genus(S_9_diatoms_wide_rel)
S_9_dinophyta_wide_rel <- combine_by_genus(S_9_dinophyta_wide_rel)

#rename tekate and atekate
rename_dino_groups <- function(df) {
  df %>%
    mutate(
      genus = case_when(
        genus == "Atekate" ~ "Peridiniales",
        genus == "Tekate" ~ "Peridiniales",
        TRUE ~ genus
      )
    )
}

R_5_dinophyta_wide_rel <- rename_dino_groups(R_5_dinophyta_wide_rel)
S_9_dinophyta_wide_rel <- rename_dino_groups(S_9_dinophyta_wide_rel)

keep_top10_and_other <- function(df) {
  num_cols <- df %>% select(where(is.numeric)) %>% names()

  df_with_total <- df %>%
    mutate(
      total_abundance = rowSums(across(all_of(num_cols)), na.rm = TRUE)
    )

  top10 <- df_with_total %>%
    arrange(desc(total_abundance)) %>%
    slice_head(n = 10)

  top10_genera <- top10$genus

  other_row <- df_with_total %>%
    filter(!genus %in% top10_genera) %>%
    summarise(
      genus = "Other",
      across(all_of(num_cols), ~ sum(.x, na.rm = TRUE))
    )
}

```

```

    )

    result <- top10 %>%
      select(-total_abundance) %>%
      bind_rows(other_row)

    result
  }
  R_5_diatoms_top10_micro <- keep_top10_and_other(R_5_diatoms_wide_rel)
  R_5_dinophyta_top10_micro <- keep_top10_and_other(R_5_dinophyta_wide_rel)
  S_9_diatoms_top10_micro <- keep_top10_and_other(S_9_diatoms_wide_rel)
  S_9_dinophyta_top10_micro <- keep_top10_and_other(S_9_dinophyta_wide_rel)

  to_long_micro <- function(df) {
    df %>%
      pivot_longer(
        cols = -genus, # all columns except genus
        names_to = "Date", # column names (dates) → this column
        values_to = "relative_abundance" # values
      )
  }

  R5_diatoms_micro_long <- to_long_micro(R_5_diatoms_top10_micro)
  R5_dino_micro_long <- to_long_micro(R_5_dinophyta_top10_micro)
  S9_diatoms_micro_long <- to_long_micro(S_9_diatoms_top10_micro)
  S9_dino_micro_long <- to_long_micro(S_9_dinophyta_top10_micro)

```

plots

metabarcoding

dinoflagellates

```

# 0) Unified Dino palette (metabarcoding + microscopy genera)

GENUS_COLORS_DINO_UNIFIED <- c(
  # Metabarcoding genera
  "Alexandrium" = "#1b9e77",
  "Apocalathium" = "#d95f02",
  "Dinophyceae_XXX" = "#e6ab02",

```

```

"Dinophysis"          = "#a6761d",
"Heterocapsa"         = "#b2df8a",
"Prorocentrum"        = "#fb9a99",
"Thoracosphaeraceae_X" = "#cab2d6",
"Tripos"              = "#fdbf6f",
"Other"               = "#bbbbbb",

# Microscopy genera (same colors as before)
"Gymnodiniales"       = "#8dd3c7",
"Oblea"               = "#ffffb3",
"Peridinales"         = "#bebada",
"Peridinium"          = "#fb8072",
"Proto-peridinium"    = "#80b1d3",
"Scrippsiella"        = "#100",

# New meta genera not in the original palettes
"Azadinium"           = "#b3de69",
"Biecheleria"         = "#fccde5",
"Karlodinium"         = "#d9d9d9",
"Pelagodinium"        = "#bc80bd"
)

## 1) Bind R-5 and S-9 metabarcoding Dinos

dino_all <- bind_rows(R5_dino_long, S9_dino_long) %>%
  mutate(
    station = factor(station_id, levels = c("R-5", "S-9")),
    Date     = as.Date(Date)
  )

## 2) Parse dates -> month_label with correct order

orders <- c("ymd", "dmy", "mdy", "Ymd", "dmy HMS", "mdy HMS", "dmy HM",
            "dmY", "Y/m/d", "d/m/Y", "b d Y", "d b Y")

df_dino_plot <- dino_all %>%
  filter(!is.na(relative_abundance)) %>%
  mutate(
    raw_Date      = as.character(Date),
    parsed_dt     = suppressWarnings(parse_date_time(raw_Date, orders = orders, exact = FALSE)),
    parsed_Date   = as_date(parsed_dt),
  )

```

```

    month_label = if_else(!is.na(parsed_Date),
                          format(parsed_Date, "%b %Y"),
                          raw_Date),
    month_order_key = if_else(!is.na(parsed_Date),
                              as.numeric(parsed_Date),
                              NA_real_)
  )

parsed_levels <- df_dino_plot %>%
  filter(!is.na(month_order_key)) %>%
  arrange(month_order_key) %>%
  pull(month_label) %>%
  unique()

unparsed_levels <- df_dino_plot %>%
  filter(is.na(month_order_key)) %>%
  pull(month_label) %>%
  unique()

all_levels <- c(parsed_levels, unparsed_levels)

df_dino_plot <- df_dino_plot %>%
  mutate(
    month_label = factor(month_label, levels = all_levels, ordered = TRUE),
    station      = factor(station, levels = c("R-5", "S-9"))
  )

## 3) Ensure "Other" is last in genus

put_other_last <- function(df) {
  gens <- df %>% distinct(genus) %>% pull(genus)
  if ("Other" %in% gens) {
    new_levels <- c(setdiff(gens, "Other"), "Other")
  } else {
    new_levels <- gens
  }
  df %>% mutate(genus = factor(genus, levels = new_levels))
}

df_dino_plot <- put_other_last(df_dino_plot)

```

```

## 4) Aggregate and (optionally) complete genus × month × station

df_dino_complete <- df_dino_plot %>%
  group_by(station, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  complete(
    station,
    month_label,
    genus,
    fill = list(relative_abundance = 0)
  ) %>%
  put_other_last() # re-apply after complete()

## 5) Subset the unified palette to the genera present in this plot

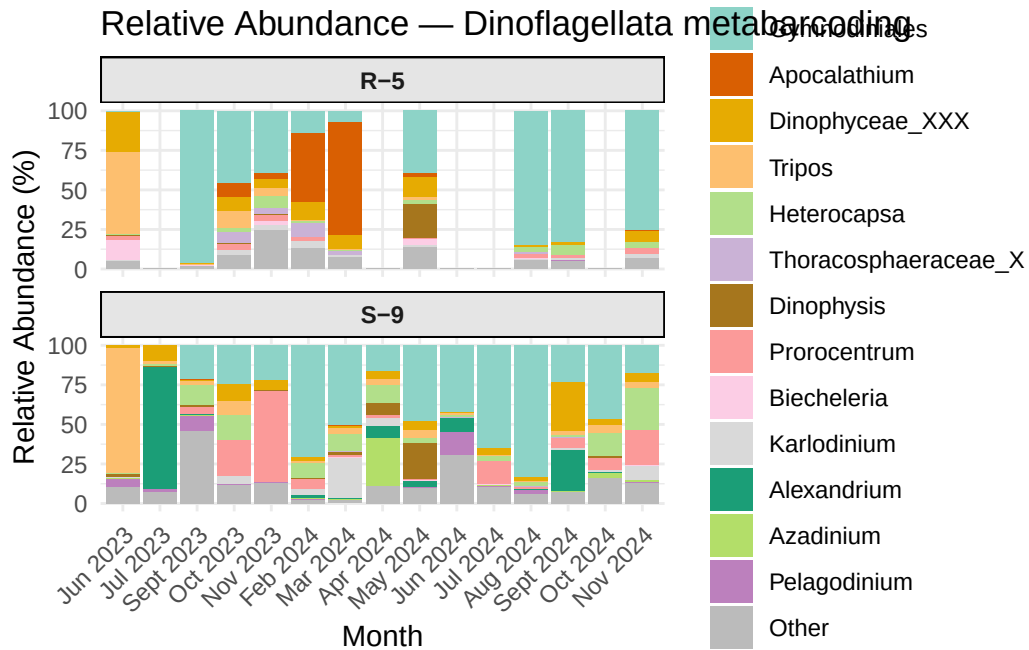
dino_colors_meta <- GENUS_COLORS_DINO_UNIFIED[levels(df_dino_complete$genus)]

## 6) Combined metabarcoding Dino plot using the new palette

combined_dino_plot <- ggplot(
  df_dino_complete,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ station, ncol = 1) +
  scale_fill_manual(values = dino_colors_meta, na.value = "grey50") +
  labs(
    title = "Relative Abundance - Dinoflagellata metabarcoding",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

```

```
print(combined_dino_plot)
```



Diatoms

```
# 0) Unified Diatom palette (meta + micro genera)
```

```
GENUS_COLORS_DIATOMS_UNIFIED <- c(
  # Metabarcoding genera
  "Acanthoceras"      = "#1b9e77",
  "Aulacoseira"       = "#d95f02",
  "Cerataulina"       = "#7570b3",
  "Chaetoceros"       = "#e7298a",
  "Cyclotella"        = "#66a61e",
  "Diatoma"           = "#e6ab02",
  "Ditylum"           = "#a6761d",
  "Leptocylindrus"    = "#666666",
  "Minidiscus"        = "#fff",
  "Other"             = "#bbbbbb",
  "Pseudo-nitzschia"  = "#b2df8a",
  "Rhizosolenia"      = "#fb9a99",
  "Skeletonema"       = "#cab2d6",
```



```

"Stephanodiscus" = "#fdbf6f",
"Thalassiosira"  = "#80b1d3",

# Microscopy genera (from your micro palette)
"Asterionellopsis" = "#1b9e77",
"Coscinodiscus"    = "#d95f02",
"Dactyliosolen"    = "#000",
"Guinardia"        = "#a6761d",
"Pseudosolenia"    = "#1f78b4",
"Sentriske"        = "#cab2d6",
"Thalassionema"   = "#fb9a99",

# New genus present in final data
"Tabellaria"       = "#8dd3c7"
)

## Helper: ensure "Other" last in genus

put_other_last <- function(df) {
  gens <- df %>% distinct(genus) %>% pull(genus)
  if ("Other" %in% gens) {
    new_levels <- c(setdiff(gens, "Other"), "Other")
  } else {
    new_levels <- gens
  }
  df %>% mutate(genus = factor(genus, levels = new_levels))
}

## 1) Combine long Diatom data

diatoms_all <- bind_rows(R5_diatoms_long, S9_diatoms_long) %>%
  mutate(
    station = factor(station_id, levels = c("R-5", "S-9"))
  )

## 2) Parse dates -> month_label with correct order

orders <- c("ymd", "dmy", "mdy", "Ymd", "dmy HMS", "mdy HMS", "dmy HM",
            "dmY", "Y/m/d", "d/m/Y", "b d Y", "d b Y")

df_diatoms_plot <- diatoms_all %>%

```

```

filter(!is.na(relative_abundance)) %>%
mutate(
  raw_Date      = as.character(Date),
  parsed_dt     = suppressWarnings(parse_date_time(raw_Date, orders = orders, exact = FALSE)),
  parsed_Date   = as_date(parsed_dt),
  month_label   = if_else(!is.na(parsed_Date),
                          format(parsed_Date, "%b %Y"),
                          raw_Date),
  month_order_key = if_else(!is.na(parsed_Date),
                            as.numeric(parsed_Date),
                            NA_real_)
)

parsed_levels <- df_diatoms_plot %>%
  filter(!is.na(month_order_key)) %>%
  arrange(month_order_key) %>%
  pull(month_label) %>%
  unique()

unparsed_levels <- df_diatoms_plot %>%
  filter(is.na(month_order_key)) %>%
  pull(month_label) %>%
  unique()

all_levels <- c(parsed_levels, unparsed_levels)

df_diatoms_plot <- df_diatoms_plot %>%
  mutate(
    month_label = factor(month_label, levels = all_levels, ordered = TRUE)
  ) %>%
  put_other_last()

## 3) Aggregate, complete grid, keep "Other" last

df_diatoms_complete <- df_diatoms_plot %>%
  group_by(station, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  complete(
    station,
    month_label,

```

```

    genus,
    fill = list(relative_abundance = 0)
  ) %>%
  put_other_last()

## 4) Subset unified palette to genera present in this plot

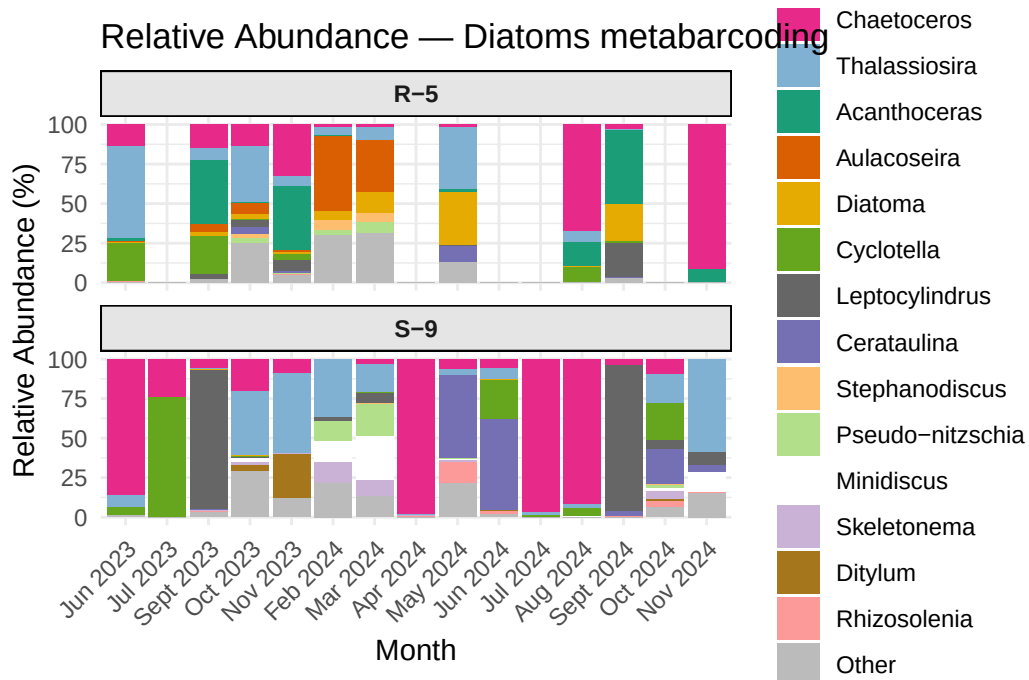
diatom_colors_meta <- GENUS_COLORS_DIATOMS_UNIFIED[levels(df_diatoms_complete$genus)]

## 5) Plot

combined_diatoms_plot <- ggplot(
  df_diatoms_complete,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ station, ncol = 1) +
  scale_fill_manual(values = diatom_colors_meta, na.value = "grey50") +
  labs(
    title = "Relative Abundance - Diatoms metabarcoding",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

print(combined_diatoms_plot)

```



microscopy

dinoflagellates

```
# 0) Unified Dino palette (same as for metabarcoding + comparisons)
```

```
GENUS_COLORS_DINO_UNIFIED <- c(
  # Metabarcoding genera
  "Alexandrium"      = "#1b9e77",
  "Apocalathium"     = "#d95f02",
  "Dinophyceae_XXX"  = "#e6ab02",
  "Dinophysis"       = "#a6761d",
  "Heterocapsa"      = "#b2df8a",
  "Prorocentrum"     = "#fb9a99",
  "Thoracosphaeraceae_X" = "#cab2d6",
  "Tripos"           = "#fdbf6f",
  "Other"            = "#bbbbbb",

  # Microscopy genera
  "Gymnodiniales"    = "#8dd3c7",
  "Oblea"            = "#ffffb3",
```

```

"Peridinales"          = "#bebada",
"Peridinium"           = "#fb8072",
"Protoperidinium"      = "#80b1d3",
"Scrippsiella" = "#000",

# New meta genera
"Azadinium"            = "#b3de69",
"Biecheleria"          = "#fccde5",
"Karlodinium"           = "#d9d9d9",
"Pelagodinium"         = "#bc80bd"
)

## Helper to put "Other" last

put_other_last <- function(df) {
  gens <- df %>% distinct(genus) %>% pull(genus)
  if ("Other" %in% gens) {
    new_levels <- c(setdiff(gens, "Other"), "Other")
  } else {
    new_levels <- gens
  }
  df %>% mutate(genus = factor(genus, levels = new_levels))
}

#### 1. Combine R-5 and S-9 and add station ####

dino_micro_all <- bind_rows(
  R5_dino_micro_long %>% mutate(station = "R-5"),
  S9_dino_micro_long %>% mutate(station = "S-9")
) %>%
  mutate(
    station = factor(station, levels = c("R-5", "S-9"))
  )

#### 2. Parse Date → month_label ####

orders <- c(
  "ymd", "dmy", "mdy", "Ymd", "dmy HMS", "mdy HMS", "dmy HM",
  "dmY", "Y/m/d", "d/m/Y", "b d Y", "d b Y"
)

```

```

df_dino_micro_plot <- dino_micro_all %>%
  filter(!is.na(relative_abundance)) %>%
  mutate(
    raw_Date      = as.character(Date),
    parsed_dt     = suppressWarnings(parse_date_time(raw_Date, orders = orders, exact = FALSE)),
    parsed_Date   = as_date(parsed_dt),
    month_label    = if_else(!is.na(parsed_Date),
                           format(parsed_Date, "%b %Y"),
                           raw_Date),
    month_order_key = if_else(!is.na(parsed_Date),
                             as.numeric(parsed_Date),
                             NA_real_)
  )

parsed_levels <- df_dino_micro_plot %>%
  filter(!is.na(month_order_key)) %>%
  arrange(month_order_key) %>%
  pull(month_label) %>%
  unique()

unparsed_levels <- df_dino_micro_plot %>%
  filter(is.na(month_order_key)) %>%
  pull(month_label) %>%
  unique()

all_levels <- c(parsed_levels, unparsed_levels)

df_dino_micro_plot <- df_dino_micro_plot %>%
  mutate(
    month_label = factor(month_label, levels = all_levels, ordered = TRUE)
  ) %>%
  put_other_last()

#### 3. Complete station × month × genus grid ####

df_dino_micro_complete <- df_dino_micro_plot %>%
  group_by(station, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  complete(
    station,

```

```

    month_label,
    genus,
    fill = list(relative_abundance = 0)
  ) %>%
  put_other_last()

#### 4. Subset unified palette to microscopy genera actually present ####

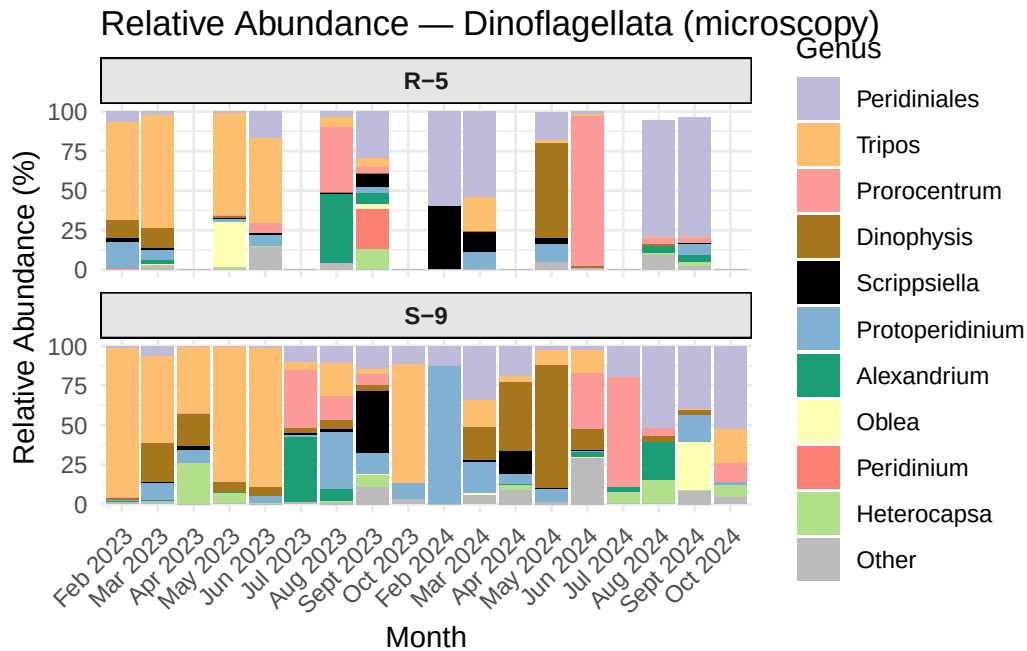
dino_colors_micro <- GENUS_COLORS_DINO_UNIFIED[levels(df_dino_micro_complete$genus)]

#### 5. Combined microscopy Dino plot ####

combined_dino_micro_plot <- ggplot(
  df_dino_micro_complete,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ station, ncol = 1) +
  scale_fill_manual(values = dino_colors_micro, na.value = "grey50") +
  labs(
    title = "Relative Abundance - Dinoflagellata (microscopy)",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

print(combined_dino_micro_plot)

```



Diatoms

0) Unified Diatom palette (same as used for metabarcoding)

```
GENUS_COLORS_DIATOMS_UNIFIED <- c(
  # Metabarcoding genera
  "Acanthoceras" = "#1b9e77",
  "Aulacoseira" = "#d95f02",
  "Cerataulina" = "#7570b3",
  "Chaetoceros" = "#e7298a",
  "Cyclotella" = "#66a61e",
  "Diatoma" = "#e6ab02",
  "Ditylum" = "#a6761d",
  "Leptocylindrus" = "#666666",
  "Minidiscus" = "#fff",
  "Other" = "#bbbbbb",
  "Pseudo-nitzschia" = "#b2df8a",
  "Rhizosolenia" = "#fb9a99",
  "Skeletonema" = "#cab2d6",
  "Stephanodiscus" = "#fdbf6f",
  "Thalassiosira" = "#80b1d3",
```



```

# Microscopy genera
"Asterionellopsis" = "#1b9e77",
"Coscinodiscus"    = "#d95f02",
"Dactyliosolen"    = "#000",
"Guinardia"        = "#a6761d",
"Pseudosolenia"    = "#1f78b4",
"Sentriske"        = "#cab2d6",
"Thalassionema"   = "#fb9a99",

# New genus in final data
"Tabellaria"       = "#8dd3c7"
)

## Helper: "Other" last

put_other_last <- function(df) {
  gens <- df %>% distinct(genus) %>% pull(genus)
  if ("Other" %in% gens) {
    new_levels <- c(setdiff(gens, "Other"), "Other")
  } else {
    new_levels <- gens
  }
  df %>% mutate(genus = factor(genus, levels = new_levels))
}

#### 1. Combine R-5 and S-9 and add station ####

diatoms_micro_all <- bind_rows(
  R5_diatoms_micro_long %>% mutate(station = "R-5"),
  S9_diatoms_micro_long %>% mutate(station = "S-9")
) %>%
  mutate(
    station = factor(station, levels = c("R-5", "S-9"))
  )

#### 2. Parse Date → month_label ####

orders <- c(
  "ymd", "dmy", "mdy", "Ymd", "dmy HMS", "mdy HMS", "dmy HM",
  "dmY", "Y/m/d", "d/m/Y", "b d Y", "d b Y"
)

```

```

df_diatoms_micro_plot <- diatoms_micro_all %>%
  filter(!is.na(relative_abundance)) %>%
  mutate(
    raw_Date = as.character(Date),
    parsed_dt = suppressWarnings(parse_date_time(raw_Date, orders = orders, exact = FALSE)),
    parsed_Date = as_date(parsed_dt),
    month_label = if_else(!is.na(parsed_Date),
                          format(parsed_Date, "%b %Y"),
                          raw_Date),
    month_order_key = if_else(!is.na(parsed_Date),
                              as.numeric(parsed_Date),
                              NA_real_)
  )

parsed_levels <- df_diatoms_micro_plot %>%
  filter(!is.na(month_order_key)) %>%
  arrange(month_order_key) %>%
  pull(month_label) %>%
  unique()

unparsed_levels <- df_diatoms_micro_plot %>%
  filter(is.na(month_order_key)) %>%
  pull(month_label) %>%
  unique()

all_levels <- c(parsed_levels, unparsed_levels)

df_diatoms_micro_plot <- df_diatoms_micro_plot %>%
  mutate(
    month_label = factor(month_label, levels = all_levels, ordered = TRUE)
  ) %>%
  put_other_last()

#### 3. Complete station × month × genus grid ####

df_diatoms_micro_complete <- df_diatoms_micro_plot %>%
  group_by(station, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  complete(
    station,

```

```

    month_label,
    genus,
    fill = list(relative_abundance = 0)
  ) %>%
  put_other_last()

#### 4. Subset palette to genera actually in the microscopy data ####

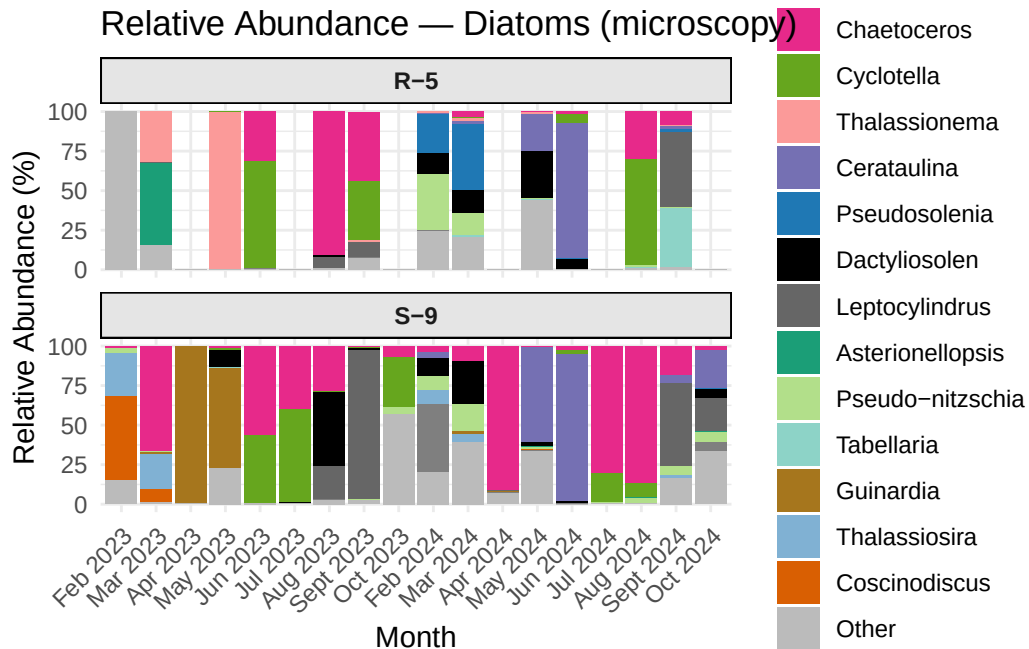
diatom_colors_micro <- GENUS_COLORS_DIATOMS_UNIFIED[levels(df_diatoms_micro_complete$genus)]

#### 5. Combined microscopy Diatom plot ####

combined_diatoms_micro_plot <- ggplot(
  df_diatoms_micro_complete,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ station, ncol = 1) +
  scale_fill_manual(values = diatom_colors_micro, na.value = "grey50") +
  labs(
    title = "Relative Abundance - Diatoms (microscopy)",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

print(combined_diatoms_micro_plot)

```



comparison

dinoflagellates R-5

```
## 1. Filter R-5 and tag datasets ####

dino_meta_R5 <- df_dino_complete %>%
  filter(station == "R-5") %>%
  mutate(
    dataset      = "Metabarcoding",
    month_label = as.character(month_label)
  )

dino_micro_R5 <- df_dino_micro_complete %>%
  filter(station == "R-5") %>%
  mutate(
    dataset      = "Microscopy",
    month_label = as.character(month_label)
  )

## 2. Keep only months where both methods have samples ####
```

```

# (non-zero abundance in at least one genus)

meta_months_nonzero <- dino_meta_R5 %>%
  group_by(month_label) %>%
  summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
  filter(total > 0) %>%
  pull(month_label)

micro_months_nonzero <- dino_micro_R5 %>%
  group_by(month_label) %>%
  summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
  filter(total > 0) %>%
  pull(month_label)

common_months <- intersect(meta_months_nonzero, micro_months_nonzero)

dino_meta_R5 <- dino_meta_R5 %>% filter(month_label %in% common_months)
dino_micro_R5 <- dino_micro_R5 %>% filter(month_label %in% common_months)

## 3. Combine and build chronological month order ####

dino_R5_both <- bind_rows(dino_meta_R5, dino_micro_R5)

all_month_labels <- unique(dino_R5_both$month_label)

month_order_tbl <- tibble(
  month_label = all_month_labels
) %>%
  mutate(
    parsed_month = parse_date_time(month_label, orders = "b Y")
  ) %>%
  filter(!is.na(parsed_month)) %>%
  arrange(parsed_month)

month_levels_ordered <- month_order_tbl$month_label

dino_R5_both <- dino_R5_both %>%
  mutate(
    month_label = factor(
      month_label,
      levels = month_levels_ordered,

```

```

    ordered = TRUE
  ),
  dataset = factor(dataset, levels = c("Metabarcoding", "Microscopy"))
)

## 4. Summarise (no complete(), only existing months) ####

dino_R5_both_sum <- dino_R5_both %>%
  group_by(dataset, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  put_other_last()

## 5. Subset unified Dino palette to genera present here ####

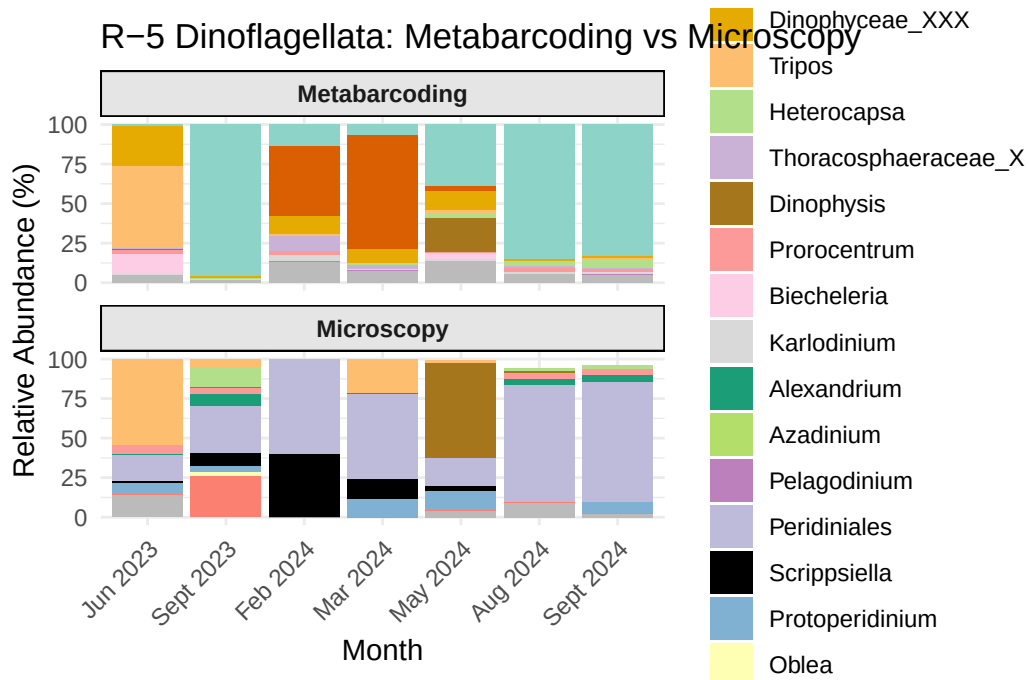
GENUS_COLORS_DINO_R5 <- GENUS_COLORS_DINO_UNIFIED[levels(dino_R5_both_sum$genus)]

## 6. Plot: Metabarcoding (top) vs Microscopy (bottom) ####

R5_dino_meta_vs_micro_plot <- ggplot(
  dino_R5_both_sum,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ dataset, ncol = 1) +
  scale_fill_manual(values = GENUS_COLORS_DINO_R5, na.value = "grey50") +
  labs(
    title = "R-5 Dinoflagellata: Metabarcoding vs Microscopy",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

print(R5_dino_meta_vs_micro_plot)

```



dinoflagellates S-9

1. Filter S-9 and make month_label character

```
dino_meta_S9 <- df_dino_complete %>%
  filter(station == "S-9") %>%
  mutate(
    dataset      = "Metabarcoding",
    month_label = as.character(month_label)
  )
```

```
dino_micro_S9 <- df_dino_micro_complete %>%
  filter(station == "S-9") %>%
  mutate(
    dataset      = "Microscopy",
    month_label = as.character(month_label)
  )
```

2. Keep only months where both methods have samples

```
meta_months_nonzero_S9 <- dino_meta_S9 %>%
```

```

group_by(month_label) %>%
  summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
  filter(total > 0) %>%
  pull(month_label)

micro_months_nonzero_S9 <- dino_micro_S9 %>%
  group_by(month_label) %>%
  summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
  filter(total > 0) %>%
  pull(month_label)

common_months_S9 <- intersect(meta_months_nonzero_S9, micro_months_nonzero_S9)

dino_meta_S9 <- dino_meta_S9 %>% filter(month_label %in% common_months_S9)
dino_micro_S9 <- dino_micro_S9 %>% filter(month_label %in% common_months_S9)

#### 3. Combine, then build a chronological month order from text ####

dino_S9_both <- bind_rows(dino_meta_S9, dino_micro_S9)

all_month_labels_S9 <- unique(dino_S9_both$month_label)

month_order_tbl_S9 <- tibble(
  month_label = all_month_labels_S9
) %>%
  mutate(
    parsed_month = parse_date_time(month_label, orders = "b Y")
  ) %>%
  filter(!is.na(parsed_month)) %>%
  arrange(parsed_month)

month_levels_ordered_S9 <- month_order_tbl_S9$month_label

dino_S9_both <- dino_S9_both %>%
  mutate(
    month_label = factor(
      month_label,
      levels = month_levels_ordered_S9,
      ordered = TRUE
    ),
    dataset = factor(dataset, levels = c("Metabarcoding", "Microscopy"))
  )

```



```

)

#### 4. Summarise without creating extra months ####

dino_S9_both_sum <- dino_S9_both %>%
  group_by(dataset, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  put_other_last()

#### 5. Shared color palette from unified Dino colors ####

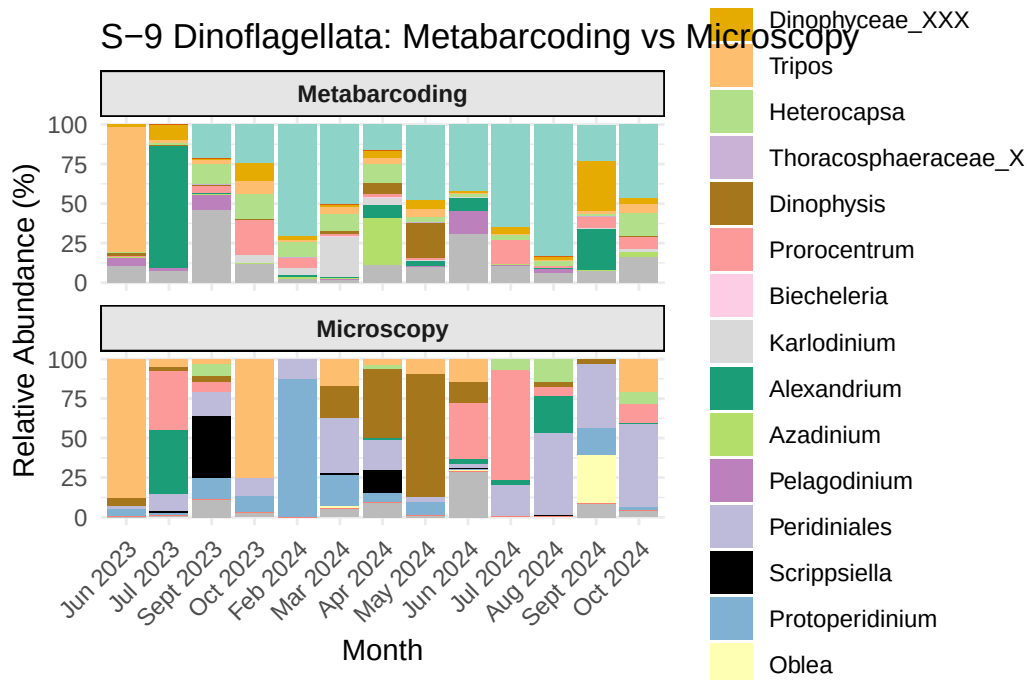
GENUS_COLORS_DINO_S9 <- GENUS_COLORS_DINO_UNIFIED[levels(dino_S9_both_sum$genus)]

#### 6. Plot: top = metabarcoding, bottom = microscopy ####

S9_dino_meta_vs_micro_plot <- ggplot(
  dino_S9_both_sum,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ dataset, ncol = 1) +
  scale_fill_manual(values = GENUS_COLORS_DINO_S9, na.value = "grey50") +
  labs(
    title = "S-9 Dinoflagellata: Metabarcoding vs Microscopy",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

print(S9_dino_meta_vs_micro_plot)

```



Diatoms R-5

1. Filter R-5 and make month_label character

```
diatoms_meta_R5 <- df_diatoms_complete %>%
  filter(station == "R-5") %>%
  mutate(
    dataset      = "Metabarcoding",
    month_label = as.character(month_label)
  )
```

```
diatoms_micro_R5 <- df_diatoms_micro_complete %>%
  filter(station == "R-5") %>%
  mutate(
    dataset      = "Microscopy",
    month_label = as.character(month_label)
  )
```

2. Keep only months where both methods have samples (non-zero)

```
meta_months_nonzero_R5_d <- diatoms_meta_R5 %>%
```

```

group_by(month_label) %>%
summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
filter(total > 0) %>%
pull(month_label)

micro_months_nonzero_R5_d <- diatoms_micro_R5 %>%
group_by(month_label) %>%
summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
filter(total > 0) %>%
pull(month_label)

common_months_R5_d <- intersect(meta_months_nonzero_R5_d, micro_months_nonzero_R5_d)

diatoms_meta_R5 <- diatoms_meta_R5 %>% filter(month_label %in% common_months_R5_d)
diatoms_micro_R5 <- diatoms_micro_R5 %>% filter(month_label %in% common_months_R5_d)

#### 3. Combine, then build a chronological month order from text ####

diatoms_R5_both <- bind_rows(diatoms_meta_R5, diatoms_micro_R5)

all_month_labels_R5_d <- unique(diatoms_R5_both$month_label)

month_order_tbl_R5_d <- tibble(
  month_label = all_month_labels_R5_d
) %>%
mutate(
  parsed_month = parse_date_time(month_label, orders = "b Y")
) %>%
filter(!is.na(parsed_month)) %>%
arrange(parsed_month)

month_levels_ordered_R5_d <- month_order_tbl_R5_d$month_label

diatoms_R5_both <- diatoms_R5_both %>%
mutate(
  month_label = factor(
    month_label,
    levels = month_levels_ordered_R5_d,
    ordered = TRUE
  ),
  dataset = factor(dataset, levels = c("Metabarcoding", "Microscopy"))

```

```

)

#### 4. Summarise without creating extra months ####

diatoms_R5_both_sum <- diatoms_R5_both %>%
  group_by(dataset, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  put_other_last()

#### 5. Shared color palette from unified diatom colors ####

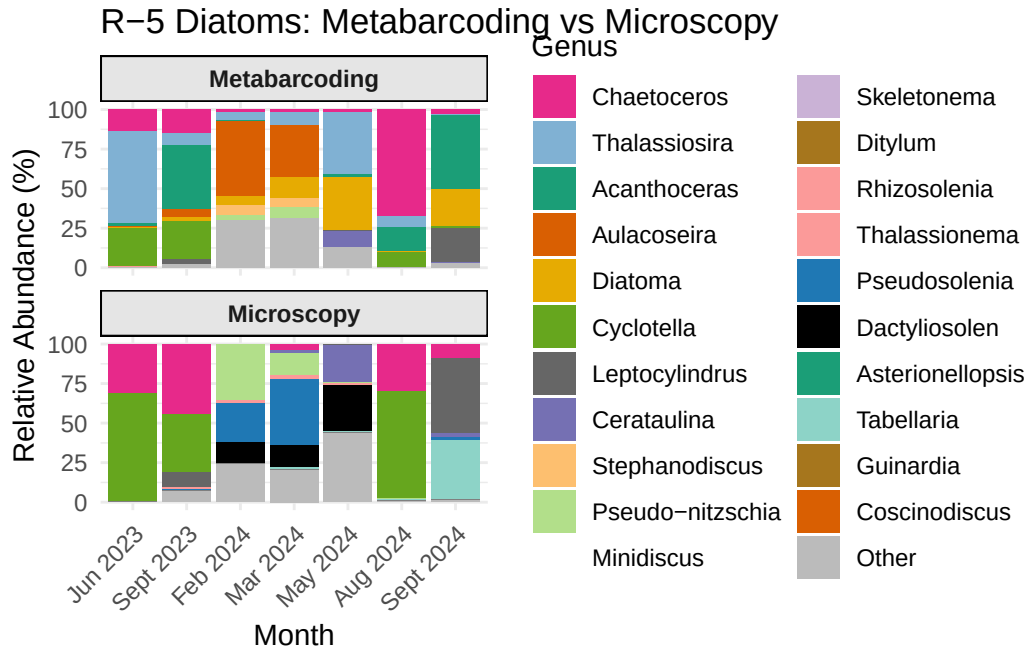
GENUS_COLORS_DIATOMS_R5 <- GENUS_COLORS_DIATOMS_UNIFIED[levels(diatoms_R5_both_sum$genus)]

#### 6. Plot: top = metabarcoding, bottom = microscopy ####

R5_diatoms_meta_vs_micro_plot <- ggplot(
  diatoms_R5_both_sum,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ dataset, ncol = 1) +
  scale_fill_manual(values = GENUS_COLORS_DIATOMS_R5, na.value = "grey50") +
  labs(
    title = "R-5 Diatoms: Metabarcoding vs Microscopy",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

print(R5_diatoms_meta_vs_micro_plot)

```



Diatoms S-9

1. Filter S-9 and make month_label character

```
diatoms_meta_S9 <- df_diatoms_complete %>%
  filter(station == "S-9") %>%
  mutate(
    dataset      = "Metabarcoding",
    month_label = as.character(month_label)
  )
```

```
diatoms_micro_S9 <- df_diatoms_micro_complete %>%
  filter(station == "S-9") %>%
  mutate(
    dataset      = "Microscopy",
    month_label = as.character(month_label)
  )
```

2. Keep only months where both methods have samples (non-zero)

```
meta_months_nonzero_S9_d <- diatoms_meta_S9 %>%
```

```

group_by(month_label) %>%
summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
filter(total > 0) %>%
pull(month_label)

micro_months_nonzero_S9_d <- diatoms_micro_S9 %>%
group_by(month_label) %>%
summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
filter(total > 0) %>%
pull(month_label)

common_months_S9_d <- intersect(meta_months_nonzero_S9_d, micro_months_nonzero_S9_d)

diatoms_meta_S9 <- diatoms_meta_S9 %>% filter(month_label %in% common_months_S9_d)
diatoms_micro_S9 <- diatoms_micro_S9 %>% filter(month_label %in% common_months_S9_d)

#### 3. Combine, then build a chronological month order from text ####

diatoms_S9_both <- bind_rows(diatoms_meta_S9, diatoms_micro_S9)

all_month_labels_S9_d <- unique(diatoms_S9_both$month_label)

month_order_tbl_S9_d <- tibble(
  month_label = all_month_labels_S9_d
) %>%
mutate(
  parsed_month = parse_date_time(month_label, orders = "b Y")
) %>%
filter(!is.na(parsed_month)) %>%
arrange(parsed_month)

month_levels_ordered_S9_d <- month_order_tbl_S9_d$month_label

diatoms_S9_both <- diatoms_S9_both %>%
mutate(
  month_label = factor(
    month_label,
    levels = month_levels_ordered_S9_d,
    ordered = TRUE
  ),
  dataset = factor(dataset, levels = c("Metabarcoding", "Microscopy"))

```

```

)

#### 4. Summarise without creating extra months ####

diatoms_S9_both_sum <- diatoms_S9_both %>%
  group_by(dataset, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
            .groups = "drop") %>%
  put_other_last()

#### 5. Shared color palette from unified diatom colors ####

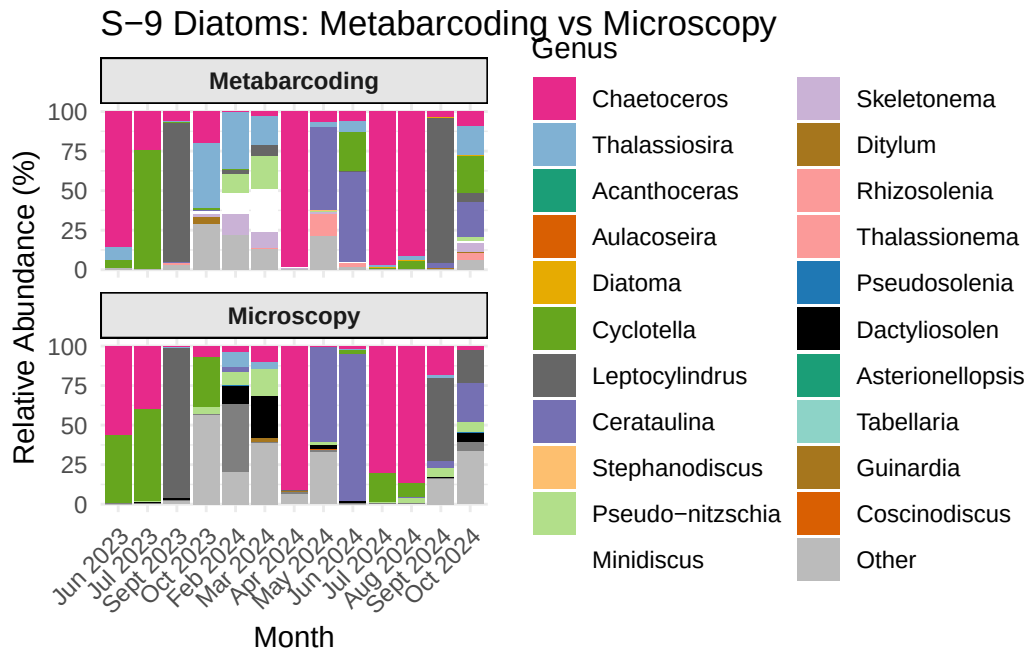
GENUS_COLORS_DIATOMS_S9 <- GENUS_COLORS_DIATOMS_UNIFIED[
  levels(diatoms_S9_both_sum$genus)
]

#### 6. Plot: top = metabarcoding, bottom = microscopy ####

S9_diatoms_meta_vs_micro_plot <- ggplot(
  diatoms_S9_both_sum,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ dataset, ncol = 1) +
  scale_fill_manual(values = GENUS_COLORS_DIATOMS_S9, na.value = "grey50") +
  labs(
    title = "S-9 Diatoms: Metabarcoding vs Microscopy",
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text       = element_text(face = "bold"),
    axis.text.x      = element_text(angle = 45, hjust = 1),
    legend.position  = "right"
  )

print(S9_diatoms_meta_vs_micro_plot)

```



Comparison new

making new datasets

```
library(dplyr)

## 1) Total per genus within each method ----

# Metabarcoding R-5 diatoms (sum across all samples)
r5_diatoms_meta_totals <- R5_diatoms_top10 %>%
  mutate(total_meta = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_meta)

# Microscopy R-5 diatoms (sum across all dates)
r5_diatoms_micro_totals <- R_5_diatoms_top10_micro %>%
  mutate(total_micro = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_micro)

## 2) Combine methods and rank genera ----

r5_diatoms_genus_rank <- full_join(
```



```

r5_diatoms_meta_totals,
r5_diatoms_micro_totals,
by = "genus"
) %>%
# genera that appear in only one table get 0 in the other
mutate(
  total_meta = replace_na(total_meta, 0),
  total_micro = replace_na(total_micro, 0),
  total_both = total_meta + total_micro
) %>%
arrange(desc(total_both))

r5_diatoms_genus_list <- union(
  R5_diatoms_top10$genus,
  R_5_diatoms_top10_micro$genus
)

## Totals per genus within each method
r5_dino_meta_totals <- R5_dino_top10 %>%
  mutate(total_meta = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_meta)

r5_dino_micro_totals <- R_5_dinophyta_top10_micro %>%
  mutate(total_micro = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_micro)

## Combine and rank
r5_dino_genus_rank <- full_join(
  r5_dino_meta_totals,
  r5_dino_micro_totals,
  by = "genus"
) %>%
mutate(
  total_meta = replace_na(total_meta, 0),
  total_micro = replace_na(total_micro, 0),
  total_both = total_meta + total_micro
) %>%
arrange(desc(total_both))

## List of genera for R-5 Dino (meta + micro)

```

```

r5_dino_genus_list <- r5_dino_genus_rank %>% pull(genus)
## Totals per genus within each method
s9_diatoms_meta_totals <- S9_diatoms_top10 %>%
  mutate(total_meta = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_meta)

s9_diatoms_micro_totals <- S9_diatoms_top10_micro %>%
  mutate(total_micro = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_micro)

## Combine and rank
s9_diatoms_genus_rank <- full_join(
  s9_diatoms_meta_totals,
  s9_diatoms_micro_totals,
  by = "genus"
) %>%
  mutate(
    total_meta = replace_na(total_meta, 0),
    total_micro = replace_na(total_micro, 0),
    total_both = total_meta + total_micro
  ) %>%
  arrange(desc(total_both))

## List of genera for S-9 Diatoms (meta + micro)
s9_diatoms_genus_list <- s9_diatoms_genus_rank %>% pull(genus)

## Totals per genus within each method
s9_dino_meta_totals <- S9_dino_top10 %>%
  mutate(total_meta = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_meta)

s9_dino_micro_totals <- S9_dinophyta_top10_micro %>%
  mutate(total_micro = rowSums(across(-genus), na.rm = TRUE)) %>%
  select(genus, total_micro)

## Combine and rank
s9_dino_genus_rank <- full_join(
  s9_dino_meta_totals,
  s9_dino_micro_totals,
  by = "genus"

```

```

) %>%
  mutate(
    total_meta = replace_na(total_meta, 0),
    total_micro = replace_na(total_micro, 0),
    total_both = total_meta + total_micro
  ) %>%
  arrange(desc(total_both))

## List of genera for S-9 Dino (meta + micro)
s9_dino_genus_list <- s9_dino_genus_rank %>% pull(genus)

collapse_to_top_genera_wide <- function(df, keep_genera) {
  keep_genera <- unique(keep_genera)

  # Ensure "Other" is included in the kept set, and not double-counted
  keep_genera_no_other <- setdiff(keep_genera, "Other")

  df_keep <- df %>%
    filter(genus %in% keep_genera_no_other)

  df_other <- df %>%
    filter(!genus %in% keep_genera_no_other)

  # If there are any "other" genera, collapse them into one "Other" row
  if (nrow(df_other) > 0) {
    other_row <- df_other %>%
      select(-genus) %>%
      summarise(across(everything(), ~ sum(.x, na.rm = TRUE))) %>%
      mutate(genus = "Other") %>%
      select(genus, everything())

    df_res <- bind_rows(df_keep, other_row)
  } else {
    df_res <- df_keep
  }

  df_res
}

r5_diatoms_genus_list <- unique(r5_diatoms_genus_list)
r5_dino_genus_list <- unique(r5_dino_genus_list)

```

```

s9_diatoms_genus_list <- unique(s9_diatoms_genus_list)
s9_dino_genus_list    <- unique(s9_dino_genus_list)

metabarcoding_R5_diatoms_top <- collapse_to_top_genera_wide(
  metabarcoding_R5_diatoms,
  keep_genera = r5_diatoms_genus_list
)
#making the datasets with the top 10 most abundant genus's at both stations

R_5_diatoms_wide_rel_top <- collapse_to_top_genera_wide(
  R_5_diatoms_wide_rel,
  keep_genera = r5_diatoms_genus_list
)
metabarcoding_R5_dino_top <- collapse_to_top_genera_wide(
  metabarcoding_R5_dino,
  keep_genera = r5_dino_genus_list
)

R_5_dinophyta_wide_rel_top <- collapse_to_top_genera_wide(
  R_5_dinophyta_wide_rel,
  keep_genera = r5_dino_genus_list
)
metabarcoding_S9_diatoms_top <- collapse_to_top_genera_wide(
  metabarcoding_S9_diatoms,
  keep_genera = s9_diatoms_genus_list
)

S_9_diatoms_wide_rel_top <- collapse_to_top_genera_wide(
  S_9_diatoms_wide_rel,
  keep_genera = s9_diatoms_genus_list
)
metabarcoding_S9_dino_top <- collapse_to_top_genera_wide(
  metabarcoding_S9_dino,
  keep_genera = s9_dino_genus_list
)

S_9_dinophyta_wide_rel_top <- collapse_to_top_genera_wide(
  S_9_dinophyta_wide_rel,
  keep_genera = s9_dino_genus_list
)

```

pivot to long

metabarcoding

```
## 1) Pivot new wide tables to long ----

to_long <- function(df) {
  df %>%
    pivot_longer(
      cols      = -genus,
      names_to  = "Sample",
      values_to = "relative_abundance"
    )
}

R5_diatoms_long <- to_long(metabarcoding_R5_diatoms_top)
R5_dino_long    <- to_long(metabarcoding_R5_dino_top)
S9_diatoms_long <- to_long(metabarcoding_S9_diatoms_top)
S9_dino_long    <- to_long(metabarcoding_S9_dino_top)

## 2) Clean metadata and extract Sample / Date / station_id ----

metadata_clean <- metadata %>%
  rename_with(.fn = ~ "Sample", .cols = matches("(?i)^file_name[_-]?r1$")) %>%
  { if (!"Sample" %in% names(.)) stop("Could not find a column matching 'file_name_R1' in
mutate(Sample = str_trim(as.character(Sample))) %>%
mutate(Sample = ifelse(str_detect(Sample, "_"),
                        str_replace(Sample, "^(.*?)_.*$", "\\1"),
                        Sample)) %>%
  {
    df <- .
    date_cols <- names(df)[str_detect(names(df), regex("date", ignore_case = TRUE))]
    has_station <- "station_id" %in% names(df)

    if (length(date_cols) == 0) {
      if (has_station) {
        df2 <- df %>% select(Sample, station_id)
      } else {
        df2 <- df %>% select(Sample)
      }
    } else {

```

```

    date_col <- date_cols[1]
    if (has_station) {
      df2 <- df %>% select(Sample, all_of(date_col), station_id) %>%
        rename(Date = all_of(date_col))
    } else {
      df2 <- df %>% select(Sample, all_of(date_col)) %>%
        rename(Date = all_of(date_col))
    }
  }
  df2
} %>%
mutate(Sample = str_trim(as.character(Sample))) %>%
filter(!(is.na(Sample) | Sample == ""))

metadata_clean <- metadata_clean %>%
  mutate(Sample = as.character(Sample),
         Sample = str_trim(Sample))

## 3) Helper: join metabarcoding long tables to metadata ----

join_with_meta <- function(mb_long, meta = metadata_clean, join_type = "left",
                           sample_col = "Sample") {

  mb <- mb_long %>%
    mutate(
      !!sample_col := as.character(.data[[sample_col]]),
      !!sample_col := str_trim(.data[[sample_col]])
    )

  joined <- switch(join_type,
    left = dplyr::left_join(mb, meta, by = sample_col),
    inner = dplyr::inner_join(mb, meta, by = sample_col),
    stop("join_type must be 'left' or 'inner'"))

  joined
}

## 4) Join each new metabarcoding long table ----

S9_diatoms_long <- join_with_meta(S9_diatoms_long, metadata_clean, join_type = "left")
R5_diatoms_long <- join_with_meta(R5_diatoms_long, metadata_clean, join_type = "left")

```

```

S9_dino_long    <- join_with_meta(S9_dino_long,    metadata_clean, join_type = "left")
R5_dino_long    <- join_with_meta(R5_dino_long,    metadata_clean, join_type = "left")

```

Microscopy

```

# Helper: pivot a wide microscopy table (genus + date columns) to long
micro_to_long <- function(df) {
  df %>%
    pivot_longer(
      cols      = -genus,          # all date columns
      names_to  = "Date",          # date as character for now
      values_to = "relative_abundance"
    )
}

# R-5
R5_diatoms_micro_long <- micro_to_long(R_5_diatoms_wide_rel_top)
R5_dino_micro_long    <- micro_to_long(R_5_dinophyta_wide_rel_top)

# S-9
S9_diatoms_micro_long <- micro_to_long(S_9_diatoms_wide_rel_top)
S9_dino_micro_long    <- micro_to_long(S_9_dinophyta_wide_rel_top)

R5_diatoms_micro_long <- R5_diatoms_micro_long %>%
  mutate(Date = dmy(Date))

R5_dino_micro_long <- R5_dino_micro_long %>%
  mutate(Date = dmy(Date))

S9_diatoms_micro_long <- S9_diatoms_micro_long %>%
  mutate(Date = dmy(Date))

S9_dino_micro_long <- S9_dino_micro_long %>%
  mutate(Date = dmy(Date))

```

Dino R-5

```
library(dplyr)
library(ggplot2)
library(lubridate)
library(tibble)

## Helper: build meta vs micro plot for one station + group -----

build_meta_micro_plot <- function(meta_long,
                                   micro_long,
                                   station_label,
                                   title,
                                   unified_palette) {
  # 1) Filter station and create month_label as character
  meta_station <- meta_long %>%
    filter(station_id == station_label) %>%
    mutate(
      dataset      = "Metabarcoding",
      month_label = format(as.Date(Date), "%b %Y")
    )

  micro_station <- micro_long %>%
    mutate(
      station      = station_label,          # ensure station info if needed
      dataset      = "Microscopy",
      month_label = format(as.Date(Date), "%b %Y")
    )

  # 2) Keep only months where both methods have non-zero abundance
  meta_months_nonzero <- meta_station %>%
    group_by(month_label) %>%
    summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
    filter(total > 0) %>%
    pull(month_label)

  micro_months_nonzero <- micro_station %>%
    group_by(month_label) %>%
    summarise(total = sum(relative_abundance, na.rm = TRUE), .groups = "drop") %>%
    filter(total > 0) %>%
    pull(month_label)
```



```

common_months <- intersect(meta_months_nonzero, micro_months_nonzero)

meta_station <- meta_station %>% filter(month_label %in% common_months)
micro_station <- micro_station %>% filter(month_label %in% common_months)

# 3) Combine and build chronological month order from text
both <- bind_rows(meta_station, micro_station)

all_month_labels <- unique(both$month_label)

month_order_tbl <- tibble(
  month_label = all_month_labels
) %>%
  mutate(
    parsed_month = parse_date_time(month_label, orders = "b Y")
  ) %>%
  filter(!is.na(parsed_month)) %>%
  arrange(parsed_month)

month_levels_ordered <- month_order_tbl$month_label

both <- both %>%
  mutate(
    month_label = factor(
      month_label,
      levels = month_levels_ordered,
      ordered = TRUE
    ),
    dataset = factor(dataset, levels = c("Metabarcoding", "Microscopy"))
  )

# 4) Summarise without creating extra months
both_sum <- both %>%
  group_by(dataset, month_label, genus) %>%
  summarise(relative_abundance = sum(relative_abundance, na.rm = TRUE),
    .groups = "drop") %>%
  put_other_last()

# 5) Shared color palette for the genera present in this plot
genus_palette <- unified_palette[levels(both_sum$genus)]

```

```

# 6) Plot
p <- ggplot(
  both_sum,
  aes(x = month_label, y = relative_abundance, fill = genus)
) +
  geom_col(position = "stack", width = 0.9) +
  facet_wrap(~ dataset, ncol = 1) +
  scale_fill_manual(values = genus_palette, na.value = "grey50") +
  labs(
    title = title,
    x = "Month",
    y = "Relative Abundance (%)",
    fill = "Genus"
  ) +
  theme_minimal() +
  theme(
    strip.background = element_rect(fill = "grey90"),
    strip.text        = element_text(face = "bold"),
    axis.text.x       = element_text(angle = 45, hjust = 1),
    legend.position   = "right"
  )

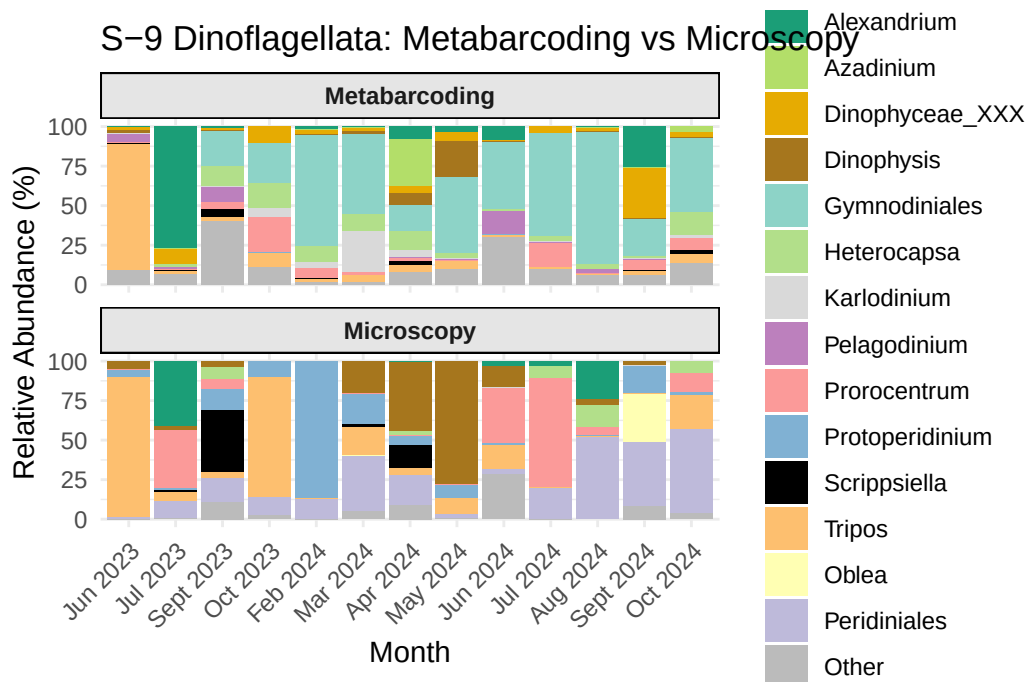
p
}

## 1) S-9 Dinoflagellates -----

S9_dino_meta_vs_micro_plot <- build_meta_micro_plot(
  meta_long      = S9_dino_long,
  micro_long     = S9_dino_micro_long,
  station_label  = "S-9",
  title          = "S-9 Dinoflagellata: Metabarcoding vs Microscopy",
  unified_palette = GENUS_COLORS_DINO_UNIFIED
)

print(S9_dino_meta_vs_micro_plot)

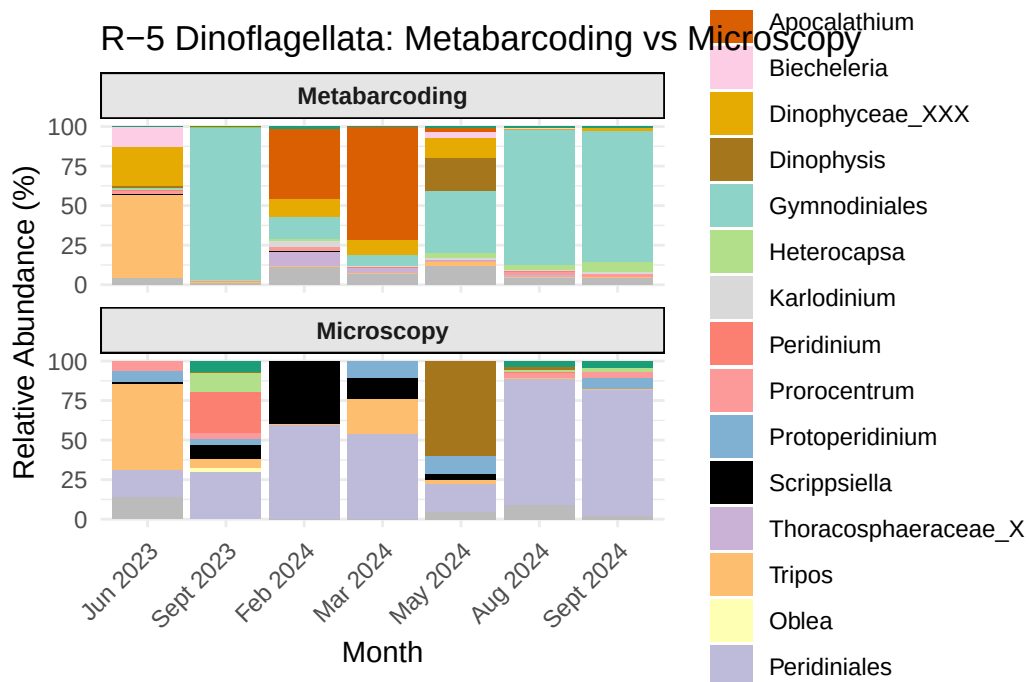
```



2) R-5 Dinoflagellates -----

```
R5_dino_meta_vs_micro_plot <- build_meta_micro_plot(
  meta_long      = R5_dino_long,
  micro_long     = R5_dino_micro_long,
  station_label  = "R-5",
  title          = "R-5 Dinoflagellata: Metabarcoding vs Microscopy",
  unified_palette = GENUS_COLORS_DINO_UNIFIED
)

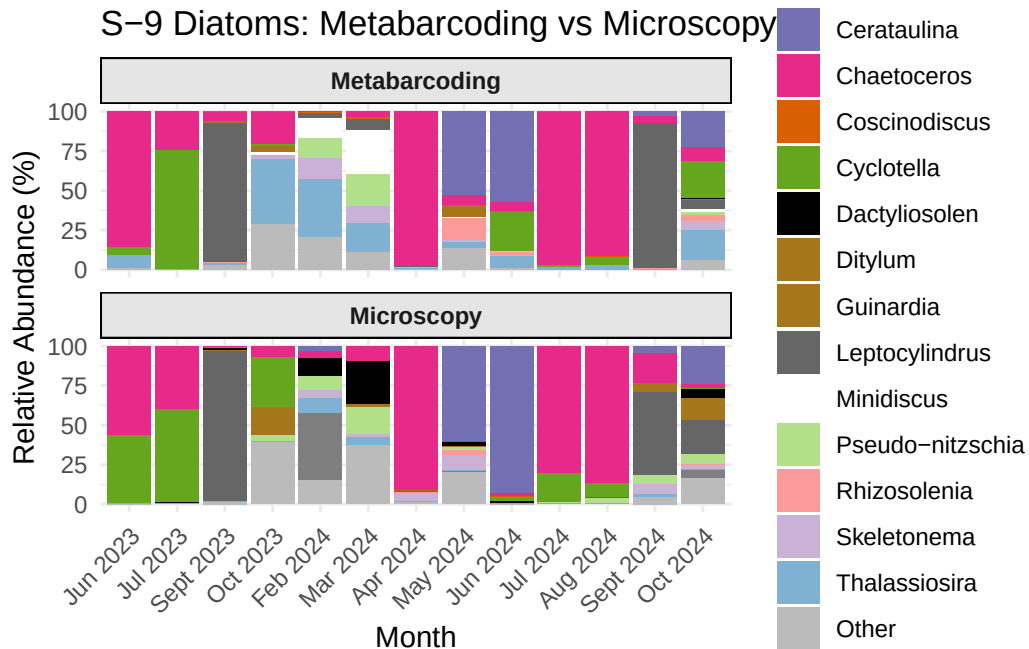
print(R5_dino_meta_vs_micro_plot)
```



3) S-9 Diatoms

```
S9_diatoms_meta_vs_micro_plot <- build_meta_micro_plot(
  meta_long      = S9_diatoms_long,
  micro_long     = S9_diatoms_micro_long,
  station_label  = "S-9",
  title          = "S-9 Diatoms: Metabarcoding vs Microscopy",
  unified_palette = GENUS_COLORS_DIATOMS_UNIFIED
)

print(S9_diatoms_meta_vs_micro_plot)
```



4) R-5 Diatoms

```
R5_diatoms_meta_vs_micro_plot <- build_meta_micro_plot(
  meta_long      = R5_diatoms_long,
  micro_long     = R5_diatoms_micro_long,
  station_label  = "R-5",
  title          = "R-5 Diatoms: Metabarcoding vs Microscopy",
  unified_palette = GENUS_COLORS_DIATOMS_UNIFIED
)

print(R5_diatoms_meta_vs_micro_plot)
```

